



FrontDesk-AI — Multi-Agent Lead-to-Call Dispatcher

multi-agent
voice routing

Yelp / Thumbtack lead → gates → first idle AI voice agent → Vapi calls in under 2 seconds

The problem

Service businesses running on Yelp and Thumbtack lose leads in seconds — the first vendor to call back wins. A staffed reception that can answer every inbound lead in under two seconds across multiple voice agents, only ringing when the area, hours, slot and TCPA checks pass, is the right shape — but it has to be both **real-time** and **reliable**.

The build

A FastAPI service with two surfaces: an **ops dashboard** (live lead feed, sub-2-second router visualisation, three-agent floor with availability pills, mock Sheet of technician slots, Vapi web-call panel), and an inbound chat console for offline scrub of the booking flow. The dispatcher runs every gate (TCPA, service area, business hours, phone present, slot available, agent rotation) and hands the call to the **first idle** AI agent (Tony, Sofia or Amanda). Vapi places the call; Claude parses the transcript into the ten booking fields after the call completes.

Architecture

```
Yelp / Thumbtack webhook
  → POST /api/leads/webhook → dispatcher gates (TCPA · area · hours · slot · rotation)
    → first idle agent in the chain (Tony · Sofia · Amanda)
      → Vapi places call (ElevenLabs voice · gpt-4o-mini · Deepgram nova-2)
  → Claude extracts the 10 booking fields from the transcript
    → Google Sheet write · real Telegram alert · retry ladder (20 min · 24 h · 72 h)
    → out-of-hours queue fires at 7:00 AM next day
```

Every gate runs with realistic wall-clock latencies (Sheets ~240 ms, geocode ~65 ms) so the demo behaves like production. Telegram alerts and the Vapi call are **real**; the Sheet and SMS are visual mocks that swap to live with a single config change.

Why it matters: sub-2-second time-to-call is the difference between winning and losing the lead; a three-agent rotation prevents one slow caller from stalling the queue; the retry ladder and out-of-hours queue convert leads the business would otherwise miss.

Stack

Python

FastAPI

Vapi

ElevenLabs

OpenAI

Deepgram

Claude

Telegram

Vercel